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A deep learning approach for mapping and dating burned areas using temporal sequences of satellite images

Miguel M. Pinto[a,*], Renata Libonati[b,a,c], Ricardo M. Trigo[a,b], Isabel F. Trigo[d], Carlos C. DaCamara[a]

[a] Instituto Dom Luiz (IDL), Faculdade de Ciências, Universidade de Lisboa, Lisbon 1749-016, Portugal
[b] Departmento de Meteorologia, Instituto de Geociências, Universidade Federal do Rio de Janeiro, Rio de Janeiro 21941-916, Brazil
[c] Centro de Estudos Florestais, Universidade de Lisboa, 1349-017 Lisboa, Portugal
[d] Departamento de Meteorologia e Geofísica, Instituto Português do Mar e da Atmosfera (IPMA), Lisbon 1749-077, Portugal

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ABSTRACT

Over the past decades, methods for burned areas mapping and dating from remote sensing imagery have been the object of extensive research. The limitations of current methods, together with the heavy pre-processing of input data they require, make them difficult to improve or apply to different satellite sensors. Here, we explore a deep learning approach based on daily sequences of multi-spectral images, as a promising and flexible technique that can be applicable to observations with various spatial and spectral resolutions. We test the proposed model for five regions around the globe using input data from VIIRS 750 m bands resampled to a 0.01° spatial resolution grid. The derived burned areas are validated against higher resolution reference maps and compared with the MCD64A1 Collection 6 and FireCCI51 global burned area datasets. We show that the proposed methodology achieves competitive results in the task of burned areas mapping, despite using lower spatial resolution observations than the two global datasets. Furthermore, we improve the task of burned areas dating for the considered regions of study when compared with state-of-the-art products. We also show that our model can be used to map burned areas for low burned fraction levels and that it can operate in near-real-time, converging to the final solution in only a few days. The obtained results are a strong indication of the advantage of deep learning approaches for the problem of mapping and dating of burned areas and provide several routes for future research.

1. Introduction

Forest fires are a natural phenomenon highly dependent on human activities, having significant impacts on social, economic and ecological levels (Patz et al., 2000, Bowman and Johnston, 2005, Driscoll et al., 2010). Mapping the fire events accurately is a problem of great importance for fire management (Bowman et al., 2011), fire danger rating and fire propagation (Freire and DaCamara, 2019, Maffei and Menenti, 2019), assessment of vegetation fire emissions and air quality (Langmann et al., 2009), land cover dynamics studies (Jin and Roy, 2005), derivation of fire patch functional traits (Laurent et al., 2018, Andela et al., 2019) and to better understand how fire patterns are evolving in a scenario of climate change and population change (Flannigan et al., 2009, Moritz et al., 2012, Andela et al., 2017).

At present, there are four main global up-to-date products for an extended period, namely: (1) the FireCCI51 (Chuvieco et al., 2018), at 250 m resolution, available for 2000–2017; (2) the MCD64A1 Collection 6 (MCD64A1C6, Giglio et al., 2018) at 500 m resolution, covering the period of 2000-present; (3) the Global Fire Emission Database (van der Werf et al., 2017), covering the period of 1997–2016 with 0.25° resolution; and (4) the FireCCILT10 (Otón and Pettinari, 2019), recently released on its beta version, covering the period of 1982–2017 also with a 0.25° resolution.

The main underlying limitation of coarse resolution products (≥250 m) is the underestimation of burned area in regions with small and highly fragmented fires (Padilla et al., 2015, Chuvieco et al., 2019), trading off with a higher temporal resolution that allows for daily date of burning assignment. Products derived from medium resolution sensors (10–80 m) have a lower temporal resolution of about 5–18 days leading to higher uncertainty in the date of burning, a problem further

* Corresponding author.
E-mail addresses: mnpinto@fc.ul.pt (M.M. Pinto), renata.libonati@igeo.ufrj.br (R. Libonati), rmtrigo@fc.ul.pt (R.M. Trigo), isabel.trigo@ipma.pt (I.F. Trigo), cdcamara@fc.ul.pt (C.C. DaCamara).

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Nomenclature

Acronym and abbreviation table

AVHRR	Advanced Very High-Resolution Radiometer
BA-Net	Burned Areas Neural Network
CEOS CalVal	Committee on Earth Observation Satellites Calibration and Validation
FireCCI	Fire Climate Change Initiative
FireCCI51	FireCCI global burned area dataset version 5.1
FireCCILT10	FireCCI Long Term dataset version 1.0
FireCCISFD11	FireCCI Small Fires Dataset version 1.1
FRP	Fire Radiative Power
ICNF	Portuguese Institute for Nature and Forest Conservation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MCD64A1C6	Collection 6 of MCD64A1 burned area product
MIR	Middle-infrared
MODIS	Moderate Resolution Imaging Spectroradiometer
MTBS	Monitoring Trends in Burn Severity
NIR	Near-infrared
NOAA	National Oceanic and Atmospheric Administration
RNNs	Recurrent Neural Networks
SLSTR	Sea and Land Surface Temperature Radiometer
TERN AusCover	Australian Terrestrial Ecosystem Research Network data delivery service
VIIRS	Visible Infrared Imaging Radiometer Suite

enhanced if clouds are present (Roteta et al., 2019). Another notable limitation of the medium and high-resolution products is the computational cost required to process those very large volumes of data, making it challenging to create and maintain an operational product at a global scale, a limitation that may be reduced with the use of distributed cloud computing systems such as Google Earth Engine (Long et al., 2019).

Several techniques to map and date burned areas using remote sensing imagery have been proposed over the years. In general terms, the procedure underlying the MCD64A1C6 and FireCCI51 algorithms consists of creating temporal composites to capture the persistent changes, filtering low-quality pixels, and then combining this information with active fires identified using the Moderate Resolution Imaging Spectroradiometer sensor (MODIS). For FireCCI51, initial burned pixels candidates are identified, and then a growing algorithm is applied to identify pixels that are likely to have burned near the initial candidates (Chuvieco et al., 2018). For MCD64A1C6 a series of steps are then applied as described in detail on Giglio et al. (2018), also including a region growing procedure, a common practice in traditional approaches (Bastarrika et al., 2011, Stroppiana et al., 2012, Alonso-Canas and Chuvieco, 2015). Other algorithms have been proposed and tested on a regional scale. For example, Pereira et al. (2017) used One-Class Support Vector Machines to reduce the omission errors caused by the omission of active fires and avoid the need of a region growing algorithm, making the procedure simpler. However, the algorithm still relies on temporal composites to address the problem of pixel contamination by clouds or cloud shadows. The use of temporal composites is a limitation as some of the information discarded could be useful. Moreover, failing to discard cloud contaminated pixels can result in unexpected behaviour as the algorithms are designed and optimized to work with clean data.

As different sensors have different characteristics, traditional burned area mapping algorithms are often designed and refined for a particular sensor, making the adaptation to another instrument often a challenging process. Furthermore, despite the significant advances of the methodologies over the years, there are still some aspects that either require improvement or are beyond the capability of the traditional approaches. In particular, the users of burned area products would benefit from longer and consistent time-series of burned areas, a better estimation of model uncertainty (Mouillot et al., 2014) and estimation of burned area fraction and combustion completeness (Chuvieco et al., 2019). It arises, therefore, the need for a methodology that is both scalable and capable of addressing these limitations. It is within this context that we would like to stress that deep learning (LeCun et al., 2015) techniques have been gaining increasing attention over the last few years. Computer vision problems, like object detection and recognition, flourished with the use of deep Convolutional Neural Networks (Krizhevsky et al., 2012) as the main building block that enables to capture the spatial correlations in image data efficiently. On the other hand, for problems with sequence data, Recurrent Neural Networks (RNNs, Rumelhart et al., 1986) are commonly used. Several types of RNNs have been proposed to better learn longer dependencies in the data. Long Short-Term Memory (LSTM, Hochreiter and Schmidhuber, 1997) have been particularly successful and are nowadays applied in numerous tasks such as speech recognition (Graves et al., 2013) and sequence to sequence models (Sutskever et al., 2014). Recent applications to remote sensing have started to emerge (Zhang et al., 2016, Zhu et al., 2017, Kemker et al., 2018, Ma et al., 2019) for land-cover classification (Hu et al., 2015, Kussul et al., 2017, Maggiori et al., 2017, Scott et al., 2017, Marcos et al., 2018), segmentation of buildings, roads and small objects (Kampffmeyer et al., 2016, Alshehhi et al., 2017), reconstruction of missing data (Zhang et al., 2018), identification of clouds and cloud shadows (Jeppesen et al., 2019, Wieland et al., 2019) and some making use of spatiotemporal structure in sequences of satellite data (Benedetti et al., 2018, Rußwurm and Körner, 2018, Pelletier et al., 2019). The problem of mapping burned areas and identifying the respective date of burning falls into the range of problems that can be addressed by these new deep learning techniques and, in turn, benefit from their inherent advantages (Reichstein et al., 2019). To our knowledge, this is the first work applying these new techniques to the problem of burned areas mapping and dating.

The approach we propose is an end-to-end model based on satellite retrieved sequences of daily images. The model uses convolutional neural networks and LSTMs with an architecture based on U-Net (Ronneberger et al., 2015). We train the model using daily top-of-atmosphere observations from Visible Infrared Imaging Radiometer Suite (VIIRS) 750 m Red, near-infrared (NIR) and middle-infrared (MIR) bands and VIIRS 375 m active fires product as inputs and using MCD64A1C6 as targets. We compare the results with MCD64A1C6 and FireCCI51 and validate against locally derived medium resolution reference maps, subject to data availability.

Our approach addresses several of the current limitations of traditional methodologies, namely: (1) No clouds/shadows mask, filtering of lower quality images or use of temporal composites is required, making the pipeline more straightforward and avoiding errors that are often observed when applying such procedures; (2) the structure of the model is not dependent on the sensor but only on the existence of input data consisting of daily sequences of images, making it an excellent candidate to be used to produce longer time-series with the resource to Advanced Very High-Resolution Radiometer (AVHRR) aboard NOAA 7–19 satellites, dating back to 1982; and (3) the incorporation of the active fires is accomplished simply by adding a “channel” to the input data without requiring any pre-processing to avoid commission errors, also opening ground for the incorporation of further input data (e.g., land-cover, topography, solar/satellite zenith angles). It is worth noticing that to the best of our knowledge this is one of the first applications of VIIRS observations to map burned areas at a global scale.

2. Methods

2.1. Study regions

Fires have different characteristics in different parts of the globe. In order to construct a diverse dataset, we selected five regions (Fig. 1) covering California, Portugal, part of the Brazilian Cerrado (hereafter referred as Brazil), Mozambique and north Australia (hereafter referred as Australia). Australia, Mozambique and Brazil are the three main choices with very different fire characteristics among them. While Australia is characterized by large fires sizes and high rate of expansion but a low density of ignitions, Mozambique has a very high density of ignitions but lower fire size and rate of expansion, and Brazil figures in between the two (Andela et al., 2019). The main motivations for including the areas of Portugal and California were the occurrence of recent tragic fires events in the past two years of 2017 (Viegas, 2018) and 2018 (Gorman et al., 2019) and the author's domain knowledge about fires in Portugal resulting from previous studies (Trigo et al., 2006, DaCamara et al., 2019) and continuous monitoring of the fires during the fire season. This domain knowledge is useful for a more insightful discussion of the case-studies shown in Section 3.4.

2.2. Datasets

Several datasets are used in this study with different usage types, namely: for model input and target; comparisons; and validation. Table 1 summarizes all the datasets used and their respective region, spatial resolution and time coverage used for this work that we now describe in the following two subsections.

2.2.1. Model inputs and targets

Daily images of Red, NIR and MIR reflectances from VIIRS Level 1B product for the period of 2012–2018, covering the five regions (http://ladsweb.modaps.eosdis.nasa.gov) were resampled to a regular grid of 0.01° spatial resolution using bilinear interpolation and selecting the image with lower satellite zenith angle if more than one image was available for a single day. Red and NIR reflectances correspond to bands M5 (centred at 0.672 μm) and M7 (centred at 0.865 μm). MIR reflectance is derived from MIR and longwave-infrared radiances (Kaufman and Remer, 1994, Libonati et al., 2010) corresponding to bands M12 (centred at 3.7 μm) and M15 (centred at 10.76 μm). VIIRS Active Fires data were obtained from https://firms.modaps.eosdis.nasa.gov for the five regions and interpolated to the working grid by summing Fire Radiative Power (FRP) values inside each grid cell for each day. Data is then organized as four-layer images with the three reflectance bands (Red, NIR and MIR) and the gridded Active Fires data.

As targets to train the model, we used MCD64A1C6, a global burned area dataset at 500 m spatial resolution, available at ftp://ba1.geog.umd.edu. This dataset includes the date of burning for each burned pixel and therefore allows to generate daily targets by resampling the original data to our working grid of 0.01° by computing the fraction of burned pixels inside each 0.01° grid cell for each day. The choice of MCD64A1C6 as the targets for train supervision instead of FireCCI51 is due to the much smaller bias in date of burning of MCD64A1C6 and the better spatial accuracy in general for the regions considered and over the 0.01° grid, when comparing with dates of active fires and medium resolution reference maps, as discussed in Section 3. We note that despite the problem being treated as classification, the targets are the burned area fractions (i.e., the fraction of area burned within each pixel) and not binary burned/non-burned masks. This can be thought as classification with soft labels.

Inputs and targets were cropped into 128x128 tiles with 28 pixels of overlap to reduce inaccuracies that can occur near the boundaries of the tiles due to the convolution operations applied by the model. The dataset covers the months of June to September for California and Portugal, July to October for Brazil and Mozambique and September to December for Australia. The selected months cover the most active period for each region (Giglio et al., 2006). This period was extended further for Portugal in 2017 (May to November) and California in 2018 (June to December), in order to include recent extreme fire events that occurred outside the usual fire season.

2.2.2. Intercomparisons and validation data

To intercompare our results, we use both MCD64A1C6, described above, and the FireCCI51 dataset, obtained from https://www.esa-fire-cci.org, that provides the date of burning for each burned pixel at 250 m resolution. As for MCD64A1C6, the FireCCI51 is resampled to the working grid considering the fraction of burned pixels inside each 0.01° grid cell for each day.

For validation, a common requirement is to use higher resolution references derived from Landsat/Sentinel imagery or other medium/high resolution source, according to the Committee on Earth Observation Satellites Calibration and Validation protocol (CEOS CalVal protocol, Boschetti et al., 2009). Due to the constraints on the availability of these burned area reference maps, several different datasets are used in this study for the validation task. For Brazil we use reference burned areas derived from 53 Landsat-8 scenes for the year of 2015, with a resolution of 30 m, matching our region of study both in time and space (Fig. 1, red patches over Brazil region). These scenes are a subset of those used by Rodrigues et al. (2019), as derived from pairs of images for nearby dates, with burned area mapped for the period between those dates, following the recommendations by the CEOS CalVal protocol. For Mozambique we use the FireCCISFD11 dataset, obtained from https://www.esa-fire-cci.org, which covers Sub-Saharan

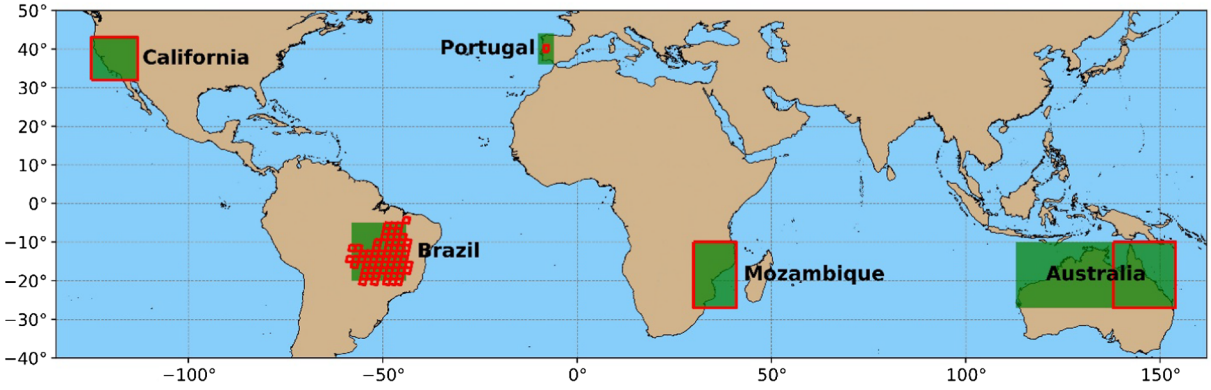


Fig. 1. Location of the 5 study regions marked by the green rectangles, corresponding to California, Portugal, Brazil, Mozambique and Australia. Red rectangles mark regions used for validation against medium resolution reference maps. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Description of the datasets used, regions covered, original resolution, used temporal coverage, and usage type.

Dataset	Region	Resolution (m)	Years	Usage
VIIRS Level 1B	All	750	2012–2018	Inputs
VIIRS Active Fires (Schroeder et al., 2014)	All	375	2012–2018	Inputs, Validation
MCD64A1C6 (Giglio et al., 2018)	All	500	2012–2018	Targets, Comparison
FireCCI51 (Chuvieco et al., 2018)	All	250	2012–2017	Comparison
Landsat-8 53 scenes (Rodrigues et al., 2019)	Brazil	30	2015	Validation
FireCCISFD11 (Roteta et al., 2019)	Mozambique	10	2016	Validation
MTBS (Eidenshink et al., 2007)	California	30 (fire perimeters)	2012–2016	Validation
TERN AusCover (Goodwin and Collett, 2014)	Australia	30	2014–2016	Validation
ICNF Burned Areas	Portugal	Variable	2017	Validation

Africa for the year of 2016 with 10 m resolution, derived from Sentinel-2 imagery. For California we use the MTBS dataset, obtained from https://www.mtbs.gov, which covers the United States of America, matching our study period on years 2012 to 2016 consisting of fire perimeters derived from 30 m resolution Landsat images. For Australia we use TERN AusCover burned areas dataset, obtained from http://www.auscover.org.au, covering north-east Australia (Fig. 1, red rectangle over the east part of Australia region) derived from Landsat imagery with 30 m resolution and matching our study period over 2012 to 2016. We considered, however, just the years 2014 to 2016 for TERN AusCover dataset, since the quality drops for the first years as evaluated visually and in comparison with MCD64A1C6 and FireCCI51. Finally, for Portugal we use a reference map obtained from ICNF (https://www.icnf.pt), corresponding to the provisory version of 5 January 2018, selecting only the fires in Central Portugal (Fig. 1, small red rectangle over Portugal region) that occurred in 2017 with source attributed to Landsat/Sentinel/Copernicus, being the most important fires mapped with resource to the medium resolution data (Landsat/Sentinel).

For the validation of the date of burning, we used VIIRS active fires because of the better detection rates of this product, compared to similar ones (Oliva and Schroeder, 2015).

All high-resolution references are interpolated for the working grid resolution of 0.01° by computing the burned area fraction inside each coarse cell.

2.3. Model

The proposed model hereafter referred to as BA-Net (from **B**urned **A**reas Neural **Net**work), is based on U-Net architecture but with several modifications in order to adapt for the problem of temporal sequences of images. First, a 3d version of U-Net is constructed based on De Fauw et al. (2018) where it is shown that separating the spatial and temporal convolutions can achieve excellent results with a much smaller and faster model. For example, a single 3d convolution with a kernel size of $7 \times 7 \times 7$ would have 343 parameters, while two convolutions, one with kernel size $1 \times 7 \times 7$, followed by another with kernel size $7 \times 1 \times 1$ only have 56 parameters. Secondly, contrary to the tasks of 3d segmentation for which 3d U-Nets are usually applied (e.g., biomedical image segmentation with 3d spatial data) the satellite data has a temporal dimension with large amounts of noise, most notably clouds. The kernel convolutions are not a natural way to learn from such noisy data, and therefore we include a unidirectional LSTM layer after the first spatial convolution, allowing the model to capture longer-term temporal dependencies in the data and to better filter the noise.

The input data consists of sequences of 64 images of size 128×128 and 4 data layers or "channels" (Red, NIR, MIR, Active Fires). Due to memory constraints, we use a batch size of 1, resulting in an input tensor with dimensions [1, 64, 4, 128, 128]. The output is also a sequence but only with one "channel" representing the probability that the pixel has burned.

Fig. 2 shows the model architecture, with the characteristic encoder/decoder (green/blue modules) of U-Nets with connections between each encoder and decoder module to avoid losing information while allowing to capture higher-level structures in the data. We now describe in detail each module of Fig. 2.

The ST-Conv3(LSTM) module (ST from **S**pace and **T**ime) consists of the layers described in Table 2 (left column). The first 3d convolution is the spatial convolution, followed by Batch Normalization and a rectified linear unit (ReLU). Then, optionally (only on the first module), an LSTM layer is used. The second 3d convolution is the temporal convolution also followed by Batch Normalization and ReLU, and finally, Dropout is applied with a probability of 0.2. On the decoder part of the model (blue modules in Fig. 2 and central column of Table 2), the UpST-Conv3 modules (Up from "**Up**ward" part of the model) have a similar structure but Transposed Convolutions are used instead, to do the upsampling. Note, however, that these modules receive two inputs, one from the module below (Fig. 2) and the other from the module at left. These inputs are concatenated over the channels dimension. The modules at the bottom of Fig. 2 (S-Conv3 and UpS-Conv3) only have the spatial convolution part, and the penultimate module named (T-Conv3) only has the temporal convolution part as suggested by the adopted terminology. The final convolution is a 3d convolution with kernel size $1 \times 1 \times 1$, used to reduce the number of channels to 1.

The exact kernel sizes and strides of the convolution layers vary from module to module and are specified in Table 3, together with the number of input and output channels and shape of the output of each

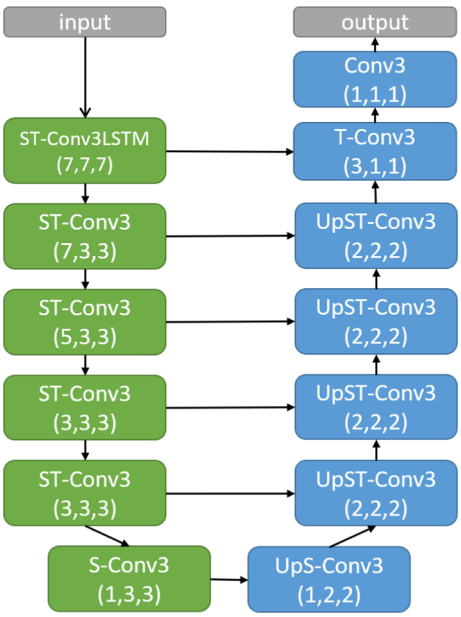


Fig. 2. Model architecture of BA-Net showing the encoder (green modules) and decoder (blue modules) sections of the model. Black arrows represent the flow of information between individual modules. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Description of the layers constituting the ST-Conv3(LSTM) and UpST-Conv3 modules.

ST-Conv3(LSTM)	UpST-Conv3	Description
Conv3d	ConvTranspose3d	3d convolution and transposed convolution (Dumoulin and Visin, 2016, Goodfellow et al., 2016)
BatchTimeNorm2d	BatchTimeNorm2d	Batch Normalization (Ioffe and Szegedy, 2015) but applied over batch and time dimensions
ReLU	ReLU	Rectified Linear Unit (Nair and Hinton, 2010)
(LSTM)	–	Long Short-Term Memory (Hochreiter and Schmidhuber, 1997)
Conv3d	ConvTranspose3d	–
BatchTimeNorm2d	BatchTimeNorm2d	–
ReLU	ReLU	–
Dropout3d	Dropout3d	3d Dropout (Srivastava et al., 2014)

module. On all convolution layers but the last, no bias parameters are used since these layers are followed by a batch normalization that has already a bias parameter. The padding on the convolution layers is set to be equal to the integer division of the respective kernel size by 2, ensuring that the output of the convolution has the same size (half the size) as the input if the stride is equal to one (equal to two). The choice of sequence size of (64, 128, 128), i.e., consisting only on powers of 2, is also to guarantee that these calculations hold.

We now describe the details of how BatchNorm and LSTM are used. Since we are using a batch size of 1, due to memory limitations, using a 3d BatchNorm would be unstable. As we are working with sequences of 64 images, we instead apply a 2d BatchNorm over batch and time dimensions together. We refer to this as BatchTimeNorm in Table 2. The LSTM layer is applied over the temporal dimension, to the individual time-series extracted from each pixel. Note that as the LSTM is placed after the spatial convolution, the time-series already capture local spatial correlations of the neighbour pixels.

As shown in Fig. 2, only the first ST-Conv3 module has an LSTM layer. In the ablation study, on Section 3.5, we show that including the LSTM layer results in an improvement of the performance when compared with an identical model without the LSTM layer. The resulting model is simple and robust, having only about 3.5 million trainable parameters. For comparison, a ResNet (He et al., 2016), a typical model for image classification, has from 11 to 60 million parameters.

2.4. Data augmentation

Data augmentation is an important aspect to consider when training a model in cases where data availability is limited to improve generalization and reduce overfitting (Perez and Wang, 2017). It is also a good way to make the model robust to specific types of noise. In real-world applications, distributions of data are usually long-tailed, meaning that some rare cases may not be sufficiently represented in the training data even if a large dataset is collected. The burned areas signature is characterized by sudden variations followed by a slow recovery. These correspond to a background pattern often covered by clouds in some regions.

To aim for better robustness and generalization, we apply the following three transformations:

1. Random brightness and contrast transformations are applied to 75% of the images. For the brightness transformation values are sampled between 0.4 and 0.6, where values bellow (above) 0.5 make the image darker (brighter). Contrast transformation is applied with a scale between 0.8 and 1.25 where values bellow (above) 1 translate to lower (higher) contrast.
2. Rectangular patches with random size and a random value, constant for the entire patch, are applied to the three first channels of the images (Red, NIR and MIR). For each image, the probability of applying patches is 0.5 with size between 5 and 50 pixels, and 1 to 5 patches, sampled from a uniform distribution.
3. Random erasing (Zhong et al., 2017) is applied with a probability of 0.5 to the Active Fires channel with size and number of patches identical to (2).

The first transformation mimics natural variability in the input images that occur due to differences in solar and satellite zenith angle. The second transformation is aimed to make the model robust to any signal that is not persistent in time, such as clouds. Finally, the random erasing of active fires is used to simulate the omission of active fires.

2.5. Implementation details

The procedure was implemented using the fastai v1.0.50.post1 (https://fast.ai) and PyTorch v1.0.1.post2 (https://pytorch.org) libraries. We train the model using One Cycle Learning (Smith, 2018) with a maximum learning rate of 0.01, Adam (Kingma and Ba, 2014) optimizer and a weight decay of 0.01. We use sigmoid activation function and Binary Cross Entropy Loss, and train the model for 8 epochs defined as 2000 individual sequences each, resulting in a total

Table 3
For each module of Fig. 2, the number of input/output channels, convolution kernel sizes and strides and size of the output sequence is described. (*for these modules the number of input channels is the double of the output channels of the previous module because those channels are concatenated with the ones coming from the encoder as shown in Fig. 2.)

Module	Input channels	Output channels	1st Conv kernel size	1st Conv stride	2nd Conv kernel size	2nd Conv stride	Output sequence size
ST-Conv3LSTM	4	32	1, 7, 7	1, 1, 1	7, 1, 1	1, 1, 1	64, 128, 128
ST-Conv3 (1)	32	64	1, 3, 3	1, 2, 2	7, 1, 1	2, 1, 1	32, 64, 64
ST-Conv3 (2)	64	128	1, 3, 3	1, 2, 2	5, 1, 1	2, 1, 1	16, 32, 32
ST-Conv3 (3)	128	256	1, 3, 3	1, 2, 2	3, 1, 1	2, 1, 1	8, 16, 16
ST-Conv3 (4)	256	256	1, 3, 3	1, 2, 2	3, 1, 1	2, 1, 1	4, 8, 8
S-Conv	256	256	1, 3, 3	1, 2, 2	–	–	4, 4, 4
UpS-Conv	256	256	1, 2, 2	1, 2, 2	–	–	4, 8, 8
UpST-Conv3 (4)	*512	256	1, 2, 2	1, 2, 2	2, 1, 1	2, 1, 1	8, 16, 16
UpST-Conv3 (3)	*512	128	1, 2, 2	1, 2, 2	2, 1, 1	2, 1, 1	16, 32, 32
UpST-Conv3 (2)	*256	64	1, 2, 2	1, 2, 2	2, 1, 1	2, 1, 1	32, 64, 64
UpST-Conv3 (1)	*128	32	1, 2, 2	1, 2, 2	2, 1, 1	2, 1, 1	64, 128, 128
T-Conv	*64	64	3, 1, 1	1, 1, 1	–	–	64, 128, 128
Conv3	64	1	1, 1, 1	1, 1, 1	–	–	64, 128, 128

train time of about 4.3 h and inference time of about 155 ms per sequence of size 64 × 4 × 128 × 128 on a NVIDIA GTX 1080 graphics card. The inference on CPU is also possible but significantly slower, with 2.7 s per sequence on an Intel Core i7-7700 CPU.

During the training phase, each sequence is sampled as follows:

1. From all images where at least 10 pixels are burned, we select one at random with reposition.
2. We select a sequence of 64 images centred at the previous selected image.
3. In case no such sequence exists (i.e., the random image is close to the beginning or end of the timeline we considered) the process is repeated until such sequence is found.

The rationale is that the model can learn faster if the samples that are given during the training are rich in burned areas.

2.6. Post-processing of model outputs

The model output sequences have the same size as the input sequences where each pixel has a value between 0 and 1 that can be interpreted as a probability that the pixel has burned. To extract monthly burned areas and date of burning, we follow the steps described below:

1. To generate the output for month m the model is provided with a sequence of 64 days starting on the 15th of the month m-1, i.e. ensuring that there are at least 15 days prior and a few more than 15 days after the month m, as represented by the 64 rectangles in Fig. 3;
2. The output is sliced to the days corresponding to the month m (green rectangles in Fig. 3) and both the sum of daily probabilities and the day the maximum probability occurs is computed for each pixel. Hereafter we refer to the sum of daily probabilities as the model confidence level;
3. The burned pixels are then given by all the pixels above a selected threshold and the date of burning is the corresponding day of the maximum probability obtained on step 2.

As a result, the first and last month for each year and each region are not used for validation.

For near real-time (NRT) mode, the approach is similar, but a sequence of 64 past images is selected instead with no slicing applied in step 2. For this case, the maximum probability and day of maximum are computed from the 64 days sequence.

2.7. Validation

In order to intercompare or validate BA-Net results with independent data not used for training, we apply Cross Validation (CV) leaving one year out each time for the training period (2012–2017), resulting in six CV folds. To evaluate the uncertainty caused by random transformations during the training of the model, for each CV fold, we train the model three times and present the average and standard deviation of the three runs, unless otherwise stated.

The spatial performance of the model is measured using Dice coefficient (also known as F_1 score) and Overall Accuracy (OA), whereas the Omission Error (OE) and Commission Error (CE) are used to calibrate the products when validating against medium resolution reference data. The metrics are defined as:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (1)$$

$$OA = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

$$OE = \frac{FN}{FN + TP} \quad (3)$$

$$CE = \frac{FP}{FP + TP} \quad (4)$$

where *TP*, *FP*, *FN* and *TN* are the number of true positives, the number of false positives, the number of false negatives and the number of true negatives, respectively.

The validation is performed over the study period for each region but discarding the first and last month of each period as extra 15 days are required prior and after the study period to apply the post-processing described in Section 2.6. For example, Mozambique region is covered from July to October, but the evaluation is done only for August and September. Hereafter, we refer to this subset as the evaluation period.

2.7.1. Intercomparisons and temporal validation

For the intercomparisons between each pair of the three products we compute the Dice coefficient, for each region, during the validation period (2012–2017), considering all the months in the evaluation period simultaneously. For the test set (2018), we compare the results of BA-Net and MCD64A1C6 with a similar approach but averaging the scores of the six CV folds. A fixed threshold of 0.5 was used to convert BA-Net confidence values to a binary burned/not burned mask, as well as for MCD64A1C6 and FireCCI51 burned area fractions on the 0.01° grid.

The dating of the burned pixels is evaluated by visual inspection of histograms of deviations between the VIIRS active fires and the date given by the product (Boschetti et al., 2010, Panisset et al., 2017, DaCamara et al., 2019), as well as by comparing the Mean Absolute Error (MAE), defined as follows:

$$MAE = \frac{1}{N}\sum_{p=1}^{N} |D_p{}^{product} - D_p{}^{reference}| \quad (5)$$

where $D_p{}^{reference}$ and $D_p{}^{product}$ are the reference and product julian days of burning for pixel p and N is the total number of pixels considered. This metric evaluates how close are the assigned dates of burning, on average, to the VIIRS active fires dates.

2.7.2. Validation with reference data and burned area fraction

As some of the reference datasets available have different temporal and spatial coverages than the five regions and periods of study (Fig. 1), the validation with reference data is restricted to pixels in the intersection of the reference data and the study spatial and temporal coverage. For Brazil however, as the reference data consists of 53 Landsat scenes covering a short period each, 2 days at the end of the period were added for FireCCI51, when matching the temporal range, in order to account for the bias in the date of burning. The value of 2 days was obtained by testing successive values and selecting the one achieving the best results.

The reference datasets used for validation for each region have different characteristics among them. Nevertheless, as emphasized in the CEOS CalVal protocol, all reference datasets consist of medium resolution data (i.e., higher resolution than the coarse products). As we are validating not only BA-Net but also FireCCI51 and MCD64A1C6, under the same conditions, any errors in the reference datasets will be common to three products and therefore the results can be compared.

As mentioned in the data section, all burned area datasets are

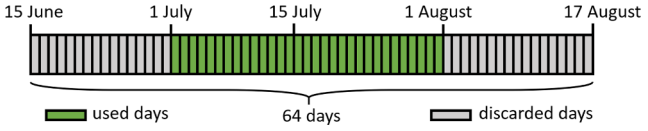


Fig. 3. Illustration of the selection of days for the monthly product for the month of July.

resampled to the 0.01° working grid by computing the fraction of burned area inside each pixel. For a general validation of the spatial burned area patterns, we consider a burned area fraction greater than or equal to 0.5 for reference maps so that a pixel is considered as burned if it has burned 50% or more, and not burned otherwise. For BA-Net, MCD64A1C6 and FireCCI51, an optimal threshold is selected based on the following procedure:

1. For the thresholds of 0.001, 0.005, 0.01 and hereafter increasing by steps of 0.01 up to 1, omission and commission errors are computed for each product and each threshold;
2. For increasing threshold values, the omission error increases and the commission error decreases. We select the threshold for which the difference between omission and commission errors is minimal, usually corresponding to the point where the two lines cross (i.e., omission error equals commission error), the exception being if the lines never cross, resulting in a saturated threshold at either 0.001 or 1;
3. Pixels with a value greater than or equal to the optimal threshold are then considered as burned and not-burned otherwise.

To evaluate the results for different reference burned area fractions a set of thresholds ranging from 0.01 up to 0.9 are used and for each of those the same procedure as before is applied to find the set of optimal thresholds for the coarse products. It should be noted that each individual reference burned area fraction threshold (from 0.01 to 0.9) results in a set of binary reference maps that are then used to find the optimal thresholds for the three coarse products.

The resulting binary maps for the three products (after applying the optimal thresholds) are evaluated against the binary reference maps using the validation metrics. This process is repeated for each one of the reference burned area fraction thresholds ranging from 0.01 to 0.9. We also note that, for each region, the months of the same year are aggregated together but different years (when available) were processed separately (i.e., optimal thresholds are computed for each year, not for each month).

3. Results and discussion

3.1. Spatial intercomparisons

Results shown in Table 4 correspond to the intercomparisons of the three burned area products for the cross-validation (2012–2017) and also the comparison between BA-Net and MCD64A1C6 for 2018 (test period). The comparisons that include BA-Net are the average of three runs with different random weight initialization and the values in square brackets the respective standard deviations. A fixed threshold of 0.5 was used to convert the BA-Net confidence values into binary burned area masks. The Dice values for 2012–2017 (Table 4) indicate that the BA-Net estimates are closer to MCD64A1C6, with scores above 0.8 for all regions but Mozambique. When comparing the Dice scores for the various pairs of products shown in Table 4, we find that matches between FireCCI51 and each of the two other products are very similar between them, as it should be expected due to the substantial similarity between BA-Net and MCD64A1C6. There are some fluctuations in the Dice values for the BA-Net - MCD64A1C6 pair obtained for the cross-validation and test periods, with some regions in the test period experiencing an increase, while others show degraded values. Such variations are to be expected since the Dice score is usually higher when large fires occur, as most of the incorrectly classified pixels are often in the fire perimeters (Libonati et al., 2015). In particular, California had several large fires in 2018, while the same year in Portugal was generally mild with a single large fire, explaining the Dice score of 0.94.

Most importantly, these results show that the proposed model learns how to map burned areas with performance level at least comparable to existing approaches using data from the VIIRS sensor resampled to about 1 × 1 km grid size. Furthermore, the methodology can be applied to data provided by any other sensor as long as its spectral channels capture the signature of wildfire in the landscape and a dataset is prepared to train the model. We also note that here we train the model from random weights initialization, however, to train on another dataset, the obtained trained weights can be used as a better initialization for a faster convergence requiring less data, a process known as transfer learning (Pan and Yang, 2009).

3.2. Temporal validation

The MAE scores were computed for each region for the cross-validation (2012–2017), and the results are presented in Table 5. BA-Net achieves the lowest MAE for all regions and an overall MAE of 1.13 days, followed by MCD64A1C6 with 1.80 days. FireCCI51 shows a high bias on the date of burning with an overall MAE of 4.25 days, more than 3 times the BA-Net score. The bias existing in FireCCI51 is due to the algorithm design. Namely, the date of burning attributed is based on the three images with the lowest NIR closest to MODIS active fires, after filtering images with low quality (Chuvieco et al., 2018). The lowest NIR is usually expected to occur after the detection of active fires and therefore burning dates attributed this way are likely to have a bias as confirmed by the results shown in the right panel of Fig. 4. Results for the test year (2018) are very similar to the cross-validation results (2012–2017). As mentioned before, the active fires are used for both input and validation of BA-Net but it is worth noting that the model was optimized to reproduce the MCD64A1C6 daily burned fractions and not the dates of the active fires. Therefore, the better performance of BA-Net on this task suggests that the model is making an appropriate use of the active fires information as it will be further discussed in the case studies of Section 3.4.

Fig. 4 shows the histograms of the difference (in days) between the detection of the active fire and the day of burn given by BA-Net, MCD64A1C6 and FireCCI51. We see that BA-Net has few negative differences, corresponding to a low rate of anticipation of the actual day of the active fire detection. The histograms confirm the overall superior performance of the BA-Net product, with very few attribution dates outside the 0–1 day range of the “true” date and matching the “true” date 70% of the cases, compared with 45% for MCD64A1C6 and just about 5% for FireCCI51. Results for MCD64A1C6 are coherent with Giglio et al. (2018) that reported a 44% match to the “true” date, globally, when comparing against MODIS active fires.

3.3. Spatial validation

The intercomparisons in Section 3.1 allowed to verify that BA-Net product is similar to MCD64A1C6 and, to a lesser extent, to FireCCI51. However, to assess the quality of a burned area product, careful validation needs to be made against higher resolution reference maps, for when they are available. In this section, we validate BA-Net,

Table 4
Dice coefficient for each region for the cross-validation (2012–2017) and test (2018) periods and every two pairs of the three products. In columns where BA-Net is present, the values correspond to the mean of 3 runs and the standard deviation is shown in square brackets.

Region	2012–2017 (cross validation)			2018 (test)
	BA-Net MCD64A1C6	BA-Net FireCCI51	MCD64A1C6 FireCCI51	BA-Net MCD64A1C6
California	0.884 [0.003]	0.831 [0.005]	0.832	0.906 [0.001]
Portugal	0.878 [0.008]	0.815 [0.010]	0.825	0.940 [0.001]
Brazil	0.831 [0.002]	0.761 [0.005]	0.753	0.780 [0.003]
Mozambique	0.742 [0.003]	0.614 [0.008]	0.626	0.749 [0.006]
Australia	0.886 [0.001]	0.749 [< 0.001]	0.735	0.881 [0.004]

Table 5
Mean absolute error between active fires and dates of burning given by BA-Net, MCD64A1C6 and FireCCI51 for the five regions and the overall score, for the cross-validation (2012–2017) and test (2018). Values for BA-Net are the average of 3 runs for cross-validation and average of the 6 CV folds for test with the standard-deviation of the 3 model runs in square brackets. Values in bold highlight the best scoring product in each case.

	2012–2017 (cross validation)			2018 (test)	
Region	BA-Net	MCD64A1C6	FireCCI51	BA-Net	MCD64A1C6
California	**0.61 [< 0.01]**	1.03	4.88	**0.70 [0.01]**	1.15
Portugal	**0.30 [< 0.01]**	0.87	4.43	**0.19 [< 0.01]**	0.43
Brazil	**0.76 [0.01]**	1.17	3.49	**0.68 [0.01]**	1.28
Mozambique	**2.02 [0.04]**	2.85	5.39	**1.82 [0.04]**	2.65
Australia	**0.39 [< 0.01]**	0.84	2.95	**0.39 [< 0.01]**	0.79
Overall	**1.13 [0.03]**	1.80	4.25	**1.11 [0.03]**	1.80

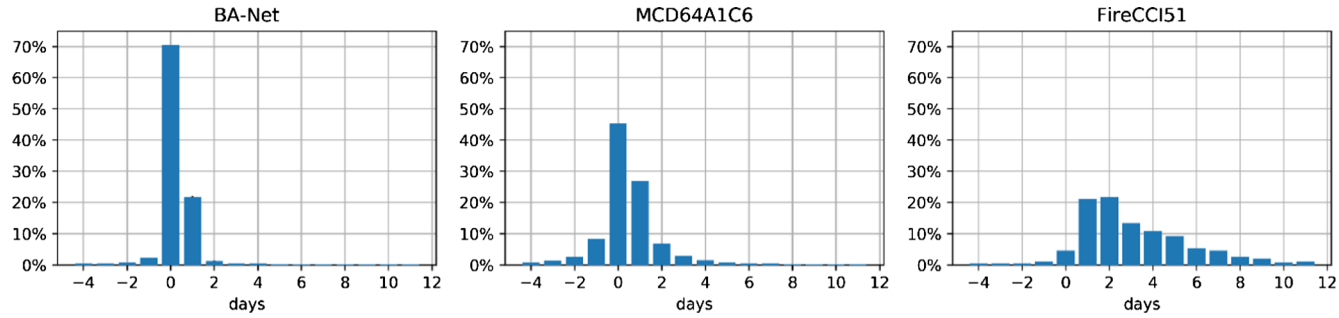


Fig. 4. Histograms of the difference in days between the predicted date of burning and detection of active fires for BA-Net, MCD64A1C6 and FireCCI51, respectively.

Table 6
Dice coefficient for validation with reference maps, considering a reference burned area fraction greater than or equal to 0.5. Columns from left to right indicate the region, the year and Dice scores for BA-Net, MCD64A1C6 and FireCCI51, respectively. BA-Net scores correspond to the mean of 3 model runs, and the standard deviation is shown in brackets. Values in bold highlight the best scoring product in each case.

Region	Year	BA-Net	MCD64A1C6	FireCCI51
Australia	2014	0.820 [< 0.001]	0.811	**0.828**
	2015	**0.729** [0.001]	0.722	0.719
	2016	**0.837** [0.014]	0.832	0.795
Mozambique	2016	0.582 [0.001]	**0.595**	0.557
California	2012	**0.920** [0.002]	0.898	0.890
	2013	**0.800** [0.002]	0.752	0.694
	2014	0.786 [0.008]	**0.791**	0.712
	2015	**0.798** [0.014]	0.791	0.657
	2016	**0.867** [0.005]	0.849	0.795
Portugal	2017	0.907 [0.001]	0.907	**0.910**
Brazil	2015	0.754 [0.003]	**0.756**	0.721
Overall		0.678 [0.002]	**0.687**	0.656

MCD64A1C6 and FireCCI51 with the independent references (as listed in Table 1, last five rows) for all the five regions. We have a particular interest in Brazil and Mozambique as they are more challenging regions due to the high density of small fires that are often not well captured by the coarse products, leading to significant underestimation of the burned area (Roteta et al., 2019). As the three coarse products are evaluated under the same conditions, concerning the same reference data, the results can then be compared to find how well BA-Net performs in relation to MCD64A1C6 and FireCCI51.

Following the procedure described in Section 2.7.2, the optimal thresholds for the three products are selected as the ones that minimize the difference between omission and commission errors for each region and year, considering a reference burned area fraction threshold of 0.5. We note that the term burned area fraction always refer to the fraction of burned area within each 0.01° size pixel. Other metrics can be used to compute the optimal thresholds and the ideal metric will depend on the intended application. As an example, if it is more important to reduce the omission error, the metric to optimize should favour a lower omission error, trading off with a higher commission error.

Table 6 shows the results corresponding to the Dice coefficient for each region and year, computed for the available data, using the optimal thresholds for each of the three products. BA-Net (Table 6, third column) outperforms MCD64A1C6 (fourth column) and FireCCI51 (last column) in Australia (over 2 of the 3 years) and California (over 4 of the 5 years). For the remaining cases BA-Net scores as the second best of the three products except on Portugal where BA-Net and MCD64A1C6 score identically and very close to FireCCI51. It is also worth noticing that BA-Net is based on VIIRS 750 m bands on a 0.01° grid, whereas MCD64A1C6 and FireCCI51 are produced at 500 and 250 m resolution, respectively, and when resampled to the 0.01° grid the burned area fraction is computed, hence keeping information from the original resolution. This difference in spatial resolution makes the task harder for BA-Net, particularly for regions with the predominance of small fires (Mozambique and Brazil), but despite that BA-Net shows competitive results for Mozambique and Brazil and outperforms MCD64A1C6 and FireCCI51 in Australia and California in general. Overall (Table 6, last row), BA-Net shows a Dice of 0.678, not far from MCD64A1C6 with a Dice of 0.687 and above FireCCI51 with a Dice of 0.656. The obtained Dice score for FireCCI51, lower than MCD64A1C6 score, is coherent with the global validation performed by Chuvieco et al. (2018) that obtained a global Dice of 0.365 for FireCCI50 (the preceding version of FireCCI51), lower than 0.478 for MCD64A1C6.

The overall results are strongly influenced by Mozambique where fires are in general small with a very high rate of ignitions, making it

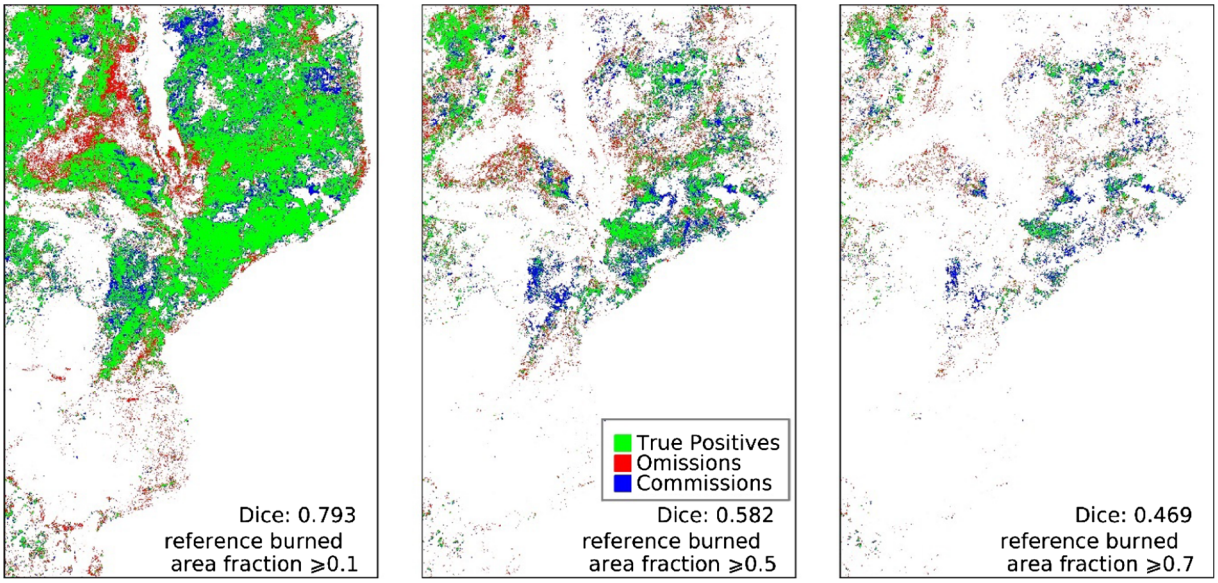


Fig. 5. Comparison of BA-Net with reference burned area fractions greater than or equal to 0.1 (left panel), 0.5 (central panel) and 0.7 (right panel) for the region of Mozambique. Green, red and blue pixels represent true positives, omissions and commissions, respectively. For BA-Net maps correspond to the average of the 3 model runs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the most challenging of the five depicted regions and making the advantage for higher resolution products more prominent. To better understand how the coarse products perform for partially burned pixels, the Dice and overall accuracy are computed for reference burned area fraction thresholds other than 0.5. Again, this makes use of the procedure described in Section 2.7.2 to find the optimal thresholds for each of the three products. We note that while for MCD64A1C6 and FireCCI51 the optimal thresholds correspond to burned area fractions, for BA-Net the optimal thresholds are based on the model confidence levels for the monthly data computed as described in Section 2.6.

Fig. 5 shows, as an example, the comparison of BA-Net with binary reference maps for reference burned area fractions greater than or equal to 0.1, 0.5 and 0.7 respectively from left to right where green, red and blue represent true positives, omission and commission. We observe that for the fraction greater than or equal to 0.1 (left panel) the number of burned pixels (i.e., the number of pixels that burned 10% or more) increases dramatically. BA-Net performs surprisingly well in this context achieving a Dice of 0.793 when using the selected optimal threshold that balances omission and commission errors.

For a more systematic evaluation of the results we compute the validation metrics for successive reference burned area fraction thresholds, ranging from 0.01 to 0.9, for each region. We note that this procedure is the same as in Fig. 5 but now considering each region and a wider set of reference burned area fraction thresholds. Results are presented in Fig. 6 in a graphical form. The Dice score, overall accuracy and optimal thresholds are shown in the left, central and right columns of Fig. 6, for each reference burned area fraction threshold and for each region – California, Portugal, Brazil, Mozambique and Australia – respectively. We note that the result obtained for each reference burned area fraction threshold can be interpreted as different classification problems: each of those cases corresponds to a (positive) classification of pixels partially burned by a fraction greater than or equal to the respective threshold.

Fig. 6 shows that, in general, BA-Net (blue line) shows Dice scores higher than MCD64A1C6 (orange line) and FireCCI51 (green line) for lower reference burned area fraction thresholds and competitive performance otherwise. The exception is the region of California where the opposite behaviour is observed, possibly explained by the different characteristics of the reference data that consists of fire perimeters (i.e., inside the fire perimeter the burned fraction is always considered to be 1). It is worth noting that the inverse U-shaped curves often observed for the Dice score (Fig. 6, left column panels) are to be expected since the task becomes harder for both lower and higher reference burned area fraction thresholds. The rationale is that the coarse resolution of the products is not sufficient to detect, as accurately, small sub-grid fires that are present in the reference maps or, conversely, to distinguish between pixels that burned almost completely.

Looking for the region of Portugal (Fig. 6, second row) the scores are very similar between the three products, BA-Net showing, however, a slightly higher overall accuracy (central panel) for reference burned area fraction thresholds below 0.5. For Mozambique, (Fig. 6, fourth row), BA-Net shows an increasingly better Dice for low burned area fraction thresholds up to 0.01 (left panel), with a clear advantage over MCD64A1C6 and FireCCI51. It should be noticed that the BA-Net Dice scores for the reference burned area fraction thresholds of 0.1, 0.5 and 0.7 in the fourth row of Fig. 6 (indicated by the three black arrows) correspond to the maps shown in Fig. 5. Contrary to the Dice score, the overall accuracy increases for higher reference burned area fraction thresholds as expected since the number of burned pixels decreases, making it easier to score well on overall but harder to have a solid match to the reference maps. The opposite is true for the lower reference burned area fraction thresholds, resulting in a drop of overall accuracy. However, this drop is very slight for BA-Net that remains with an overall accuracy above 0.85, even for a reference burned area fraction threshold of 0.01, contrasting to the two other products that show an accuracy below 0.75. Fig. 6 third row shows the results for Brazil, corresponding to the overall score for the 53 Landsat scenes. We see that BA-Net have a similar Dice as MCD64A1C6 but surpassing it for reference burned area fraction thresholds below about 0.35, reinforcing the observation that BA-Net has a better performance for small fires. For Australia (Fig. 6, last row), BA-Net outperforms MCD64A1C6 and FireCCI51 for reference burned area fraction thresholds below about 0.6 with a higher Dice score and a higher overall accuracy for reference burned area fraction thresholds between about 0.15 and 0.6.

The general improvement of BA-Net for small fractions can have a substantial impact for the estimation of the total burned area in regions where small fires prevail, and most importantly, BA-Net achieves overall competitive results in burned area mapping, even when compared with products, such as MCD64A1C6 and FireCCI51, that have the advantage of using higher spatial resolution observations. The choice of VIIRS 750 m

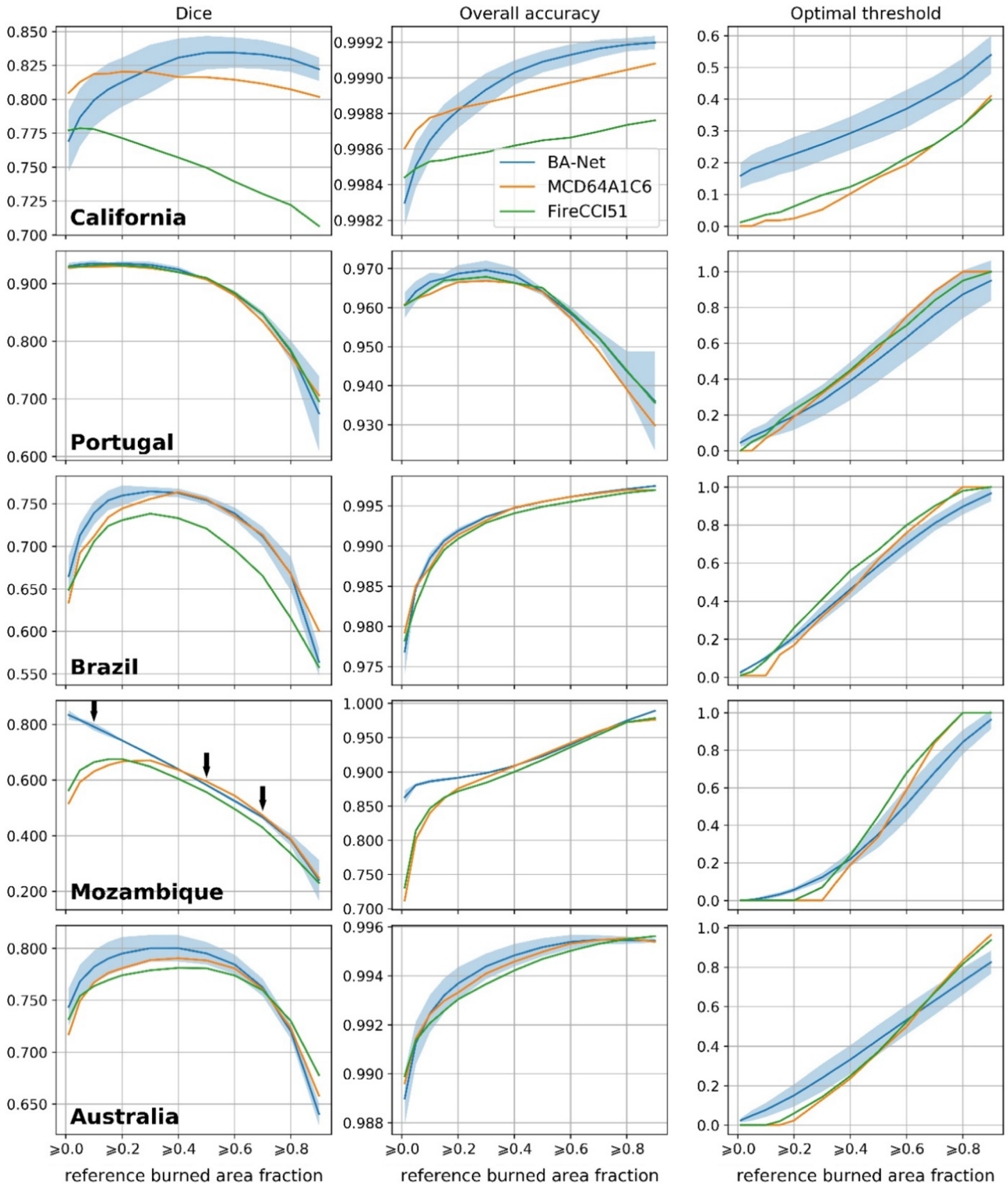


Fig. 6. Dice, overall accuracy and optimal thresholds (left, middle, and right panels, respectively) for each reference burned area fraction for the BA-Net (blue), MCD64A1C6 (orange) and FireCCI51 (green), for the regions of California, Portugal, Brazil, Mozambique and Australia (first to last row). The shaded blue region for BA-Net corresponds to the 95% confidence interval of 3 model runs. The three black arrows indicate the Dice values corresponding to the thresholds showed in Fig. 5 (0.1, 0.5 and 0.7). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

bands instead of VIIRS 375 m bands is necessary for this discussion since otherwise it could be argued that a better dataset could be the cause of BA-Net good results. However, by using the 750 m resampled to 0.01° grid we show competitive and state-of-the-art results despite the lower resolution, corroborating the benefit of BA-Net and the deep learning framework for the problem of mapping and dating of burned areas.

3.4. Case studies and near real-time application

To better understand the BA-Net model performance, we show case studies for two fires that occurred in Portugal. The first occurred on 15/16 October 2017 in central Portugal (Pinto et al., 2018) and the second on 3–9 August 2018 in Algarve, southern Portugal (Rego et al., 2019).

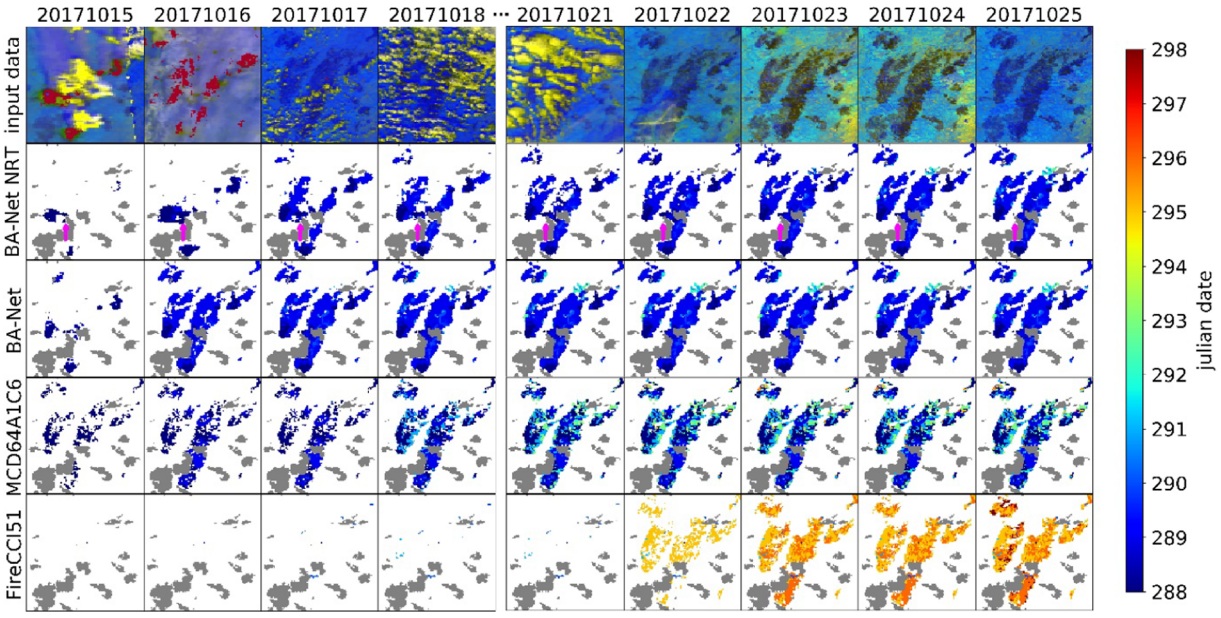


Fig. 7. An example of 11 consecutive days, for 15 (day 288) to 25 (day 298) October for Central Portugal covering a region of about 1 × 1°. The lines from top to bottom correspond to: (1) crop of a sequence of input images represented with RGB colours corresponding to bands Red, NIR, MIR inverted and active fires overlaid in red; (2) BA-Net burned area (coloured surface) and date of burning (according to the colorbar) in NRT mode; (3) as in (2) but in monthly mode; (4) burned area (coloured surface) and date of burning (according to the colorbar) for the MCD64A1C6; (5) as in (4) but for FireCCI51. Shaded grey regions in rows 2–5 correspond to areas burned in the same year but previous to the displayed time window. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

These two case studies have very distinct characteristics, which allow us to gain insight into the inner-working of BA-Net.

Fig. 7 shows the sequence of input images (first row) corresponding to 15 to 25 October (19–20 October are not shown as VIIRS images are completely covered by clouds) in Central Portugal where the colours correspond to an RGB composite with bands Red, NIR and MIR inverted, and the active fires overlapped in red. The second and third rows (Fig. 7) show the BA-Net in near real-time (NRT) mode (i.e., only considering input data up to the event date shown on top of each panel) and in monthly mode (i.e., allowing input data after the event date), while the last two sequences show MCD64A1C6 and FireCCI51, respectively. This particular case study is very challenging, as the presence of clouds during the fire event did not allow a clear view of the surface during the first few days. Good quality images, where burned areas can be accurately identified, are available only one week after the fire events (on 23 and 24 October).

We first note (Fig. 7) that BA-Net and MCD64A1C6 have similar characteristics, whereas FireCCI51 has a higher bias for the date of burning as mentioned before. Another important aspect is that active fires on the 15 October (Fig. 7; red dots in the 1st top panel) extend outside the burned area, resulting in commission errors appearing on BA-Net NRT on 15/16 October, as pointed by the magenta arrow on the second row. However, as the model is exposed to new images with better quality, particularly starting on 22 October, it updates the outputs according to the new information, converging rapidly to the final solution shown in the third row. This result is relevant, since if the BA-Net estimates were too influenced by (overfitting to) the active fires, then that whole region where active fires are present should be identified as burned. The evolution of the BA-Net burned areas suggests that instead, the model is able to reach a balance between the various inputs. Furthermore, the evolution of the BA-Net NRT (second row) shows that for 16 October only a few pixels are classified as burned.

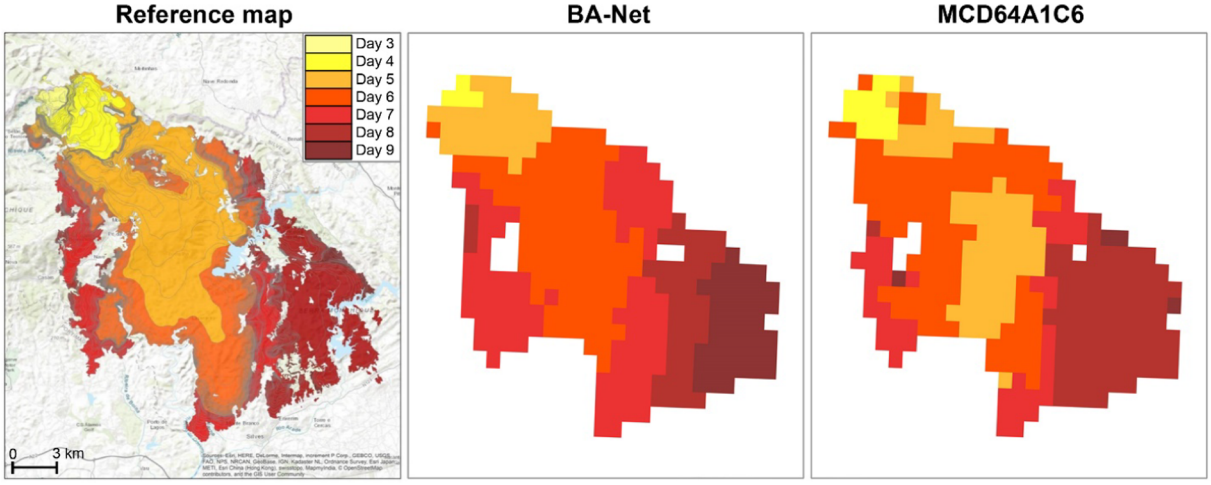


Fig. 8. Reference burned area and dates of burning (left panel, adapted from Rego et al., 2019), BA-Net mapping (center panel) and MCD64A1C6 mapping (right panel). All dates of burning are coloured according to the color legend on the left panel.

However, as we move forward in time, it includes more burned regions and maps the date of burning to the correct or close to the correct day, suggesting that the model "understands" the spatiotemporal correlations between the active fires and the reflectances leading to more accurate and spatially coherent dates of burning.

Looking now for the case study of August 2018, Fig. 8 shows the comparison between BA-Net and MCD64A1C6 final dates of burning and a high-resolution analysis adapted from a local evaluation report for this fire event (Rego et al., 2019). We can see that BA-Net dates of burning (central panel) follow a pattern close to the "true" fire progression (left panel), whereas MCD64A1C6 (right panel) shows a less realistic fire progression. These small differences can have a substantial impact for applications such as the derivation of fire patch functional traits (Laurent et al., 2018, Andela et al., 2019) such as daily fire expansion speed, direction and fire line length. The one-day delay well visible between the reference map and BA-Net is a consequence of a different definition of day as the day for BA-Net corresponds to the period between two satellite observations. In the Monchique fire, the period of higher activity and higher burned area expansion occurred mostly in the afternoon, after the satellite passage, hence the apparent delay of 1-day from the real day.

3.5. Ablation study

In this section, we test the impact of the main model design choices on the results, namely the LSTM layer and the U-Net structure. For these experiments, we train the model with the years of 2012–2017 but leaving 2015 out for validation and we report the results for 2015. This is enough to test the differences in performance using intercomparisons (Table 7) with MCD64A1C6 and validation results for Australia, California and Brazil for the year 2015 (Table 8) to show that the decrease in the similarity between BA-Net and MCD64A1C6 also translates to differences with the reference data.

Table 7 shows that removing the LSTM or using a shallower model without the U-Net structure (i.e., following the direct path input → ST-Conv3LSTM → T-Conv3 → Conv3 → output and removing all remaining modules) results in an overall decrease in performance as measured by the Dice coefficient, a result that is corroborated by the results in Table 8. The exception being the region of California where the performance increases when compared with the reference data. This could, however, be a consequence of the different characteristics of the reference data for California as discussed before.

When using the shallower model and removing the LSTM layer simultaneously results drop further (Table 7 last row, Table 8 last column), corroborating the observation that LSTM layer has an important impact on the model performance.

4. Conclusions

Systematic mapping and dating of burned areas is a task of great importance for a wide range of applications, such as fire management, calibration of fire propagation models, assessment of vegetation fire emissions and air quality, land cover dynamics studies, derivation of fire patch functional traits and to better understand how the fire patterns are evolving in a scenario of climate change and population change.

Here we proposed a model capable of learning to identify and assign dates to burned areas from sequences of multi-spectral daily images retrieved from satellite without the need for cloud masking or image selection as pre-processing. The trained model provides competitive results on burned area mapping, for a reference threshold of 0.5, showing the second best overall Dice of 0.678, not far from MCD64A1C6 0.687 and better than FireCCI51 with 0.656. For lower burned area fraction thresholds, BA-Net shows generally higher Dice scores than the two other products, except for the region of California where it shows instead higher Dice score for higher burned area fraction thresholds. Such behaviour is likely to be caused by the different characteristics of the reference for this region, as discussed in Section 3.3.

For the task of dating of fire events, BA-Net shows clearly state-of-the-art results with an overall mean absolute error of 1.13 days, followed by MCD64A1C6 with 1.80 and FireCCI51 with 4.25 when validating for the 6 cross-validation years (2012–2017) using dates of the active fires as reference. The two case studies analysed in Section 3.4 corroborate these results and further show that BA-Net assigned dates of burning are consistent in space, even for difficult cases such as the fires of October in Portugal, contrary to what is observed for MCD64A1C6 and FireCCI51. The better consistency of dates of burning is a consequence of BA-Net model structure that is able to capture the spatiotemporal relations of the daily surface reflectances and active fires.

The case studies in Section 3.4 also show that BA-Net NRT converges rapidly to the final monthly product result with a rate that depends primarily on cloud coverage. The NRT product has important applications for emergency services and quick assessments of burned area. In this line, the European Forest Fire Information System (San-Miguel-Ayanz et al., 2012) has a Rapid Damage Assessment module that produces near real-time burned area maps using, however, a semi-automatic approach, that requires manual verification and correction. As BA-Net outputs converge to the final solution in a few days, we expect it to improve on this task by reducing significantly the amount of manual work needed to map the burned areas and by providing more accurate dates of burning.

In recent years deep learning has been driving important progress in many different areas, including data-driven Earth system science (Pelletier et al., 2019, Reichstein et al., 2019). Such an achievement is only possible due to the better capacity of these algorithms to extract meaningful information from large amounts of data in a computationally feasible way (Reichstein et al., 2019). Mapping and dating of burned areas with satellite retrieved imagery is the type of problem that can benefit from recent deep learning advances. The results obtained in this work are a first step towards bringing these great advances in computer vision of the recent few years to the domain of burned areas mapping with satellite imagery, a challenging problem that has been the subject of research for more than four decades (Hitchcock and Hoffer, 1974). The good results obtained in this work and the prospects of a continuous increase in computational power and data availability suggest that this is a path that is worth following in years to come, given the high potential for future applications.

Despite the good results obtained we acknowledge that there are,

Table 7
Effects on model performance (baseline) of removing the LSTM layer (w/o LSTM), removing the U-Net architecture, i.e. following a direct path input → ST-Conv3LSTM → T-Conv3 → Conv3 → output (w/o U-Net arch) and removing both the previous two together (w/o LSTM, U-Net arch). Values for each region correspond to the Dice coefficient for the year 2015, using MCD64A1C6 as reference. Values in bold highlight the best scoring model for each region.

Model/Region	California	Portugal	Brazil	Mozambique	Australia
baseline	**0.832 [0.011]**	**0.846 [0.003]**	**0.819 [0.001]**	**0.727 [0.013]**	**0.885 [0.001]**
w/o LSTM	0.815 [0.010]	0.808 [0.002]	0.786 [0.005]	0.656 [0.005]	0.866 [0.002]
w/o U-Net	0.794 [0.003]	0.807 [0.008]	0.796 [0.001]	0.714 [0.010]	0.860 [< 0.001]
w/o U-Net and LSTM	0.740 [0.004]	0.780 [0.014]	0.748 [0.003]	0.652 [0.002]	0.797 [0.003]

Table 8
Similar to Table 7 but using the higher resolution reference maps as reference.

Region	Year	baseline	w/o LSTM	w/o U-Net	w/o U-Net and LSTM
Australia	2015	**0.729 [0.001]**	0.716 [0.003]	0.710 [0.001]	0.686 [0.003]
California	2015	0.798 [0.014]	**0.847 [0.002]**	0.842 [0.001]	0.826 [0.008]
Brazil	2015	**0.754 [0.003]**	0.734 [0.002]	0.735 [0.001]	0.712 [0.002]

however, several open questions. In the short-term we plan to investigate how the model performs in other regions, namely over higher latitudes, by studying burned areas in boreal forests and tundra, both covering about 33% of the global land surface (MCGuire et al., 1995, Young et al., 2017). This is an important step towards the application of this methodology to derive a global burned area product, either from VIIRS or other sensor with similar bands onboard a polar-orbiter (e.g., SLSTR onboard Sentinel-3). Similarly, we expect to test the BA-Net to coarser sensors such as AVHRR, which in turn have the potential to provide longer time-series, as well as to medium resolution sensors (10–80 m) such as Landsat-8 and Sentinel-2. The need of different thresholds for different regions/fire regimes is also a question we plan to address in future work, namely by incorporating input information suitable to explain the differences in fire regimes, such as land cover and topography. Sociocultural factors also play an important role in fire ignition patterns that may be possible to incorporate indirectly based on historical fire data.

With the aim of accelerating research on these and many other topics that may benefit from the methodology presented in this work, code, pre-trained models and datasets used to train the model are available at https://github.com/mnpinto/banet.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Alonso-Canas, I., Chuvieco, E., 2015. Global burned area mapping from ENVISAT-MERIS and MODIS active fire data. Remote Sens. Environ. 163, 140–152. https://doi.org/10.1016/j.rse.2015.03.011.

Alshehhi, R., Marpu, P.R., Woon, W.L. Dalla, Mura, M., 2017. Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks. ISPRS J. Photogramm. Remote Sens. 130, 139–149. https://doi.org/10.1016/j.isprsjprs.2017.05.002.

Andela, N., Morton, D.C., Giglio, L., Chen, Y., Van Der Werf, G.R., Kasibhatla, P.S., DeFries, R.S., Collatz, G.J., Hantson, S., Kloster, D., Bachelet, M., Forrest, G., Lasslop, F., Li, S., Mangeon, J.R., Melton, C., Yue, C., Randerson, D., 2017. A human-driven decline in global burned area. Science 356 (6345), 1356–1362. https://doi.org/10.1126/science.aal4108.

Andela, N., Morton, D.C., Giglio, L., Paugam, R., Chen, Y., Hantson, S., van der Werf, G.R., Randerson, J.T., 2019. The Global Fire Atlas of individual fire size, duration, speed and direction. Earth Syst. Sci. Data 11 (2), 529–552. https://doi.org/10.5194/essd-11-529-2019.

Bastarrika, A., Chuvieco, E., Martín, M.P., 2011. Mapping burned areas from Landsat TM/ETM+ data with a two-phase algorithm: Balancing omission and commission errors. Remote Sens. Environ. 115 (4), 1003–1012. https://doi.org/10.1016/j.rse.2010.12.005.

Benedetti, P., Ienco, D., Gaetano, R., Ose, K., Pensa, R.G., Dupuy, S., 2018. M3Fusion: a deep learning architecture for multiscale multimodal multitemporal satellite data fusion. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 11 (12), 4939–4949. https://doi.org/10.1109/JSTARS.2018.2876357.

Boschetti, L., Roy, D.P., Justice, C.O., 2009. International Global Burned Area Satellite Product Validation Protocol. Part I – production and standardization of validation reference data, https://lpvs.gsfc.nasa.gov/PDF/BurnedAreaValidationProtocol.pdf (accessed 22 June, 2019).

Boschetti, L., Roy, D.P., Justice, C.O., Giglio, L., 2010. Global assessment of the temporal reporting accuracy and precision of the MODIS burned area product. Int. J. Wildl. Fire 19 (6), 705–709. https://doi.org/10.1071/WF09138.

Bowman, D.M.J.S., Balch, J., Artaxo, P., Bond, W.J., Cochrane, M.A., D'Antonio, C.M., DeFries, R., Johnston, F.H., Keeley, J.E., Krawchuk, M.A., Kull, C.A., Mack, M., Moritz, M.A., Pyne, S., Roos, C.I., Scott, A.C., Sodhi, N.S., Swetnam, T.W., Whittaker, R., 2011. The human dimension of fire regimes on Earth. J. Biogeogr. 38 (12), 2223–2236. https://doi.org/10.1111/j.1365-2699.2011.02595.x.

Bowman, D.M.J.S., Johnston, F.H., 2005. Wildfire smoke, fire management, and human health. EcoHealth 2 (1), 76–80. https://doi.org/10.1007/s10393-004-0149-8.

Chuvieco, E., Lizundia-Loiola, J., Pettinari, M.L., Ramo, R., Padilla, M., Tansey, K., Mouillot, F., Laurent, P., Storm, T., Heil, A., Plummer, S., 2018. Generation and analysis of a new global burned area product based on MODIS 250 m reflectance bands and thermal anomalies. Earth Syst. Sci. Data 10 (4), 2015–2031. https://doi.org/10.5194/essd-10-2015-2018.

Chuvieco, E., Mouillot, F., van der Werf, G.R., San Miguel, J., Tanasse, M., Koutsias, N., García, M., Yebra, M., Padilla, M., Gitas, I., Heil, A., Hawbaker, T.J., Giglio, L., 2019. Historical background and current developments for mapping burned area from satellite Earth observation. Remote Sens. Environ. 225, 45–64. https://doi.org/10.1016/j.rse.2019.02.013.

DaCamara, C.C., Libonati, R., Pinto, M.M., Hurduc, A., 2019. Near-and middle-infrared monitoring of burned areas from space. in satellite information classification and interpretation. IntechOpen. https://doi.org/10.5772/intechopen.82444.

De Fauw, J., Ledsam, J.R., Romera-Paredes, B., Nikolov, S., Tomasev, N., Blackwell, S., Askham, H., Glorot, X., O'Donoghue, B., Visentin, D., van den Driessche, G., Lakshminarayanan, B., Meyer, C., Mackinder, F., Bouton, S., Ayoub, K., Chopra, R., King, D., Karthikesalingam, A., Hughes, C.O., Raine, R., Hughes, J., Sim, D.A., Egan, C., Tufail, A., Montgomery, H., Hassabis, D., Rees, G., Back, T., Khaw, P.T., Suleyman, M., Cornebise, J., Keane, P.A., Ronneberger, O., 2018. Clinically applicable deep learning for diagnosis and referral in retinal disease. Nat. Med. 24 (9), 1342. https://doi.org/10.1038/s41591-018-0107-6.

Driscoll, D.A., Lindenmayer, D.B., Bennett, A.F., Bode, M., Bradstock, R.A., Cary, G.J., Clarke, M.F., Dexter, N., Fensham, R., Friend, G., Gill, M., James, S., Kay, G., Keith, D.A., MacGregor, C., Russell-Smith, J., Salt, D., Watson James, J.E.M., Williams, R.J., York, A., 2010. Fire management for biodiversity conservation: key research questions and our capacity to answer them. Biol. Conserv. 143 (9), 1928–1939. https://doi.org/10.1016/j.biocon.2010.05.026.

Dumoulin, V., Visin, F., 2016. A guide to convolution arithmetic for deep learning. arXiv preprint arXiv:1603.07285.

Eidenshink, J., Schwind, B., Brewer, K., Zhu, Z.L., Quayle, B., Howard, S., 2007. A project for monitoring trends in burn severity. Fire ecology 3 (1), 3–21. https://doi.org/10.4996/fireecology.0301003.

Flannigan, M.D., Krawchuk, M.A., de Groot, W.J., Wotton, B.M., Gowman, L.M., 2009. Implications of changing climate for global wildland fire. Int. J. Wildl. Fire 18 (5), 483–507. https://doi.org/10.1071/WF08187.

Freire, J.G., DaCamara, C.C., 2019. Using cellular automata to simulate wildfire propagation and to assist in fire management. Nat. Hazards Earth Syst. Sci. 19 (1), 169–179. https://doi.org/10.5194/nhess-19-169-2019.

Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. Remote Sens. Environ. 217, 72–85. https://doi.org/10.1016/j.rse.2018.08.005.

Giglio, L., Van der Werf, G.R., Randerson, J.T., Collatz, G.J., Kasibhatla, P., 2006. Global estimation of burned area using MODIS active fire observations. Atmos. Chem. Phys. 6 (4), 957–974. https://doi.org/10.5194/acp-6-957-2006.

Goodfellow, I., Bengio, Y., Courville, A., 2016. Deep Learning. MIT Press. https://www.deeplearningbook.org.

Goodwin, N.R., Collett, L.J., 2014. Development of an automated method for mapping fire history captured in Landsat TM and ETM+ time series across Queensland, Australia. Remote Sens. Environ. 148, 206–221. https://doi.org/10.1016/j.rse.2014.03.021.

Gorman, C., Feng, Y., Chambers, J., Stapp, J., 2019. Camp fire processed landsat 8 images, pre-fire, during-fire, post-fire. environmental system science data infrastructure for a virtual ecosystem. Next-Gen. Ecosyst. Experim. (NGEE) Trop. https://doi.org/10.15485/1512511.

Graves, A., Mohamed, A.R., Hinton, G., 2013. Speech recognition with deep recurrent neural networks. In: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, Vancouver, BC, Canada, 26–31 May, pp. 6645–6649. https://doi.org/10.1109/ICASSP.2013.6638947.

He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 27–30 June, pp. 770–778. https://doi.org/10.1109/CVPR.2016.90.
Hitchcock, H.C., Hoffer, R.M., 1974. Mapping a recent forest fire with ERTS-1 MSS data. 3rd Remote sensing of earth resources. In: Third Conference on Earth and Information Analysis System. Tullahoma, TN, 25–27 March, vol. 3. pp. 449–461.
Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9 (8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735.
Hu, F., Xia, G.S., Hu, J., Zhang, L., 2015. Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery. Remote Sens. 7 (11), 14680–14707. https://doi.org/10.3390/rs71114680.
Ioffe, S., Szegedy, C., 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift. arXiv preprint arXiv:1502.03167.
Jin, Y., Roy, D.P., 2005. Fire-induced albedo change and its radiative forcing at the surface in northern Australia. Geophys. Res. Lett. 32 (13). https://doi.org/10.1029/2005GL022822.
Jeppesen, J.H., Jacobsen, R.H., Inceoglu, F., Toftegaard, T.S., 2019. A cloud detection algorithm for satellite imagery based on deep learning. Remote Sens. Environ. 229, 247–259. https://doi.org/10.1016/j.rse.2019.03.039.
Kampffmeyer, M., Salberg, A.B., Jenssen, R., 2016. Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, Las Vegas, NV, USA, 26 June–1 July, pp. 1–9. https://doi.org/10.1109/CVPRW.2016.90.
Kaufman, Y.J., Remer, L.A., 1994. Detection of forests using mid-IR reflectance: an application for aerosol studies. IEEE Trans. Geosci. Remote Sens. 32 (3), 672–683. https://doi.org/10.1109/36.29798.
Kemker, R., Salvaggio, C., Kanan, C., 2018. Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning. ISPRS J. Photogramm. Remote Sens. 145, 60–77. https://doi.org/10.1016/j.isprsjprs.2018.04.014.
Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.
Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. In: Advances in Neural Information Processing Systems, Lake Tahoe, Nevada, 3–6 December, pp. 1097–1105.
Kussul, N., Lavreniuk, M., Skakun, S., Shelestov, A., 2017. Deep learning classification of land cover and crop types using remote sensing data. IEEE Geosci. Remote Sens. Lett. 14 (5), 778–782. https://doi.org/10.1109/LGRS.2017.2681128.
Langmann, B., Duncan, B., Textor, C., Trentmann, J., van der Werf, G.R., 2009. Vegetation fire emissions and their impact on air pollution and climate. Atmos. Environ. 43 (1), 107–116. https://doi.org/10.1016/j.atmosenv.2008.09.047.
Laurent, P., Mouillot, F., Yue, C., Ciais, P., Moreno, M.V., Nogueira, J.M., 2018. FRY, a global database of fire patch functional traits derived from space-borne burned area products. Sci. Data 5, 180132. https://doi.org/10.1038/sdata.2018.132.
LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. Nature 521 (7553), 436. https://doi.org/10.1038/nature14539.
Libonati, R., DaCamara, C.C., Pereira, J.M.C., Peres, L.F., 2010. Retrieving middle-infrared reflectance for burned area mapping in tropical environments using MODIS. Remote Sens. Environ. 114 (4), 831–843. https://doi.org/10.1016/j.rse.2009.11.018.
Libonati, R., DaCamara, C.C., Setzer, A.W., Morelli, F., Melchiori, A.E., 2015. An algorithm for burned area detection in the Brazilian Cerrado using 4 μm MODIS imagery. Remote Sens. 7 (11), 15782–15803. https://doi.org/10.3390/rs71115782.
Long, T., Zhang, Z., He, G., Jiao, W., Tang, C., Wu, B., Zhang, X., Wang, G., Yin, R., 2019. 30 m resolution global annual burned area mapping based on landsat images and google earth engine. Remote Sens. 11 (5), 489. https://doi.org/10.3390/rs11050489.
Ma, L., Liu, Y., Zhang, X., Ye, Y., Yin, G., Johnson, B.A., 2019. Deep learning in remote sensing applications: a meta-analysis and review. ISPRS J. Photogramm. Remote Sens. 152, 166–177. https://doi.org/10.1016/j.isprsjprs.2019.04.015.
Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. ISPRS J. Photogramm. Remote Sens. 158, 263–278. https://doi.org/10.1016/j.isprsjprs.2019.10.013.
Maggiori, E., Tarabalka, Y., Charpiat, G., Alliez, P., 2017. Convolutional neural networks for large-scale remote-sensing image classification. IEEE Trans. Geosci. Remote Sens. 55 (2), 645–657. https://doi.org/10.1109/TGRS.2016.2612821.
Marcos, D., Volpi, M., Kellenberger, B., Tuia, D., 2018. Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models. ISPRS J. Photogramm. Remote Sens. 145, 96–107. https://doi.org/10.1016/j.isprsjprs.2018.01.021.
McGuire, A.D., Melillo, J.M., Kicklighter, D.W., Joyce, L.A., 1995. Equilibrium responses of soil carbon to climate change: empirical and process-based estimates. J. Biogeogr. 785–796. https://doi.org/10.2307/2845980.
Moritz, M.A., Parisien, M.A., Batllori, E., Krawchuk, M.A., Van Dorn, J., Ganz, D.J., Hayhoe, K., 2012. Climate change and disruptions to global fire activity. Ecosphere 3 (6), 1–22. https://doi.org/10.1890/ES11-00345.1.
Mouillot, F., Schultz, M.G., Yue, C., Cadule, P., Tansey, K., Ciais, P., Chuvieco, E., 2014. Ten years of global burned area products from spaceborne remote sensing—a review: analysis of user needs and recommendations for future developments. Int. J. Appl. Earth Obs. Geoinf. 26, 64–79. https://doi.org/10.1016/j.jag.2013.05.014.
Nair, V., Hinton, G.E., 2010. Rectified linear units improve restricted boltzmann machines. In: ICML-10 Proceedings of the 27th International Conference on Machine Learning, Haifa, Israel, 21–24 June, pp. 807–814.
Oliva, P., Schroeder, W., 2015. Assessment of VIIRS 375 m active fire detection product for direct burned area mapping. Remote Sens. Environ. 160, 144–155. https://doi.org/10.1016/j.rse.2015.01.010.
Otón, G., Pettinari, M.L., 2019. ESA CCI ECV Fire Disturbance: D3.3.4 Product User Guide – LTDR, version 1.0. Available at: http://www.esa-fire-cci.org/documents (accessed 22 June, 2019).
Padilla, M., Stehman, S.V., Ramo, R., Corti, D., Hantson, S., Oliva, P., Alonso-Canas, I., Bradley, A.V., Tansey, K., Mota, B., Pereira, J.M., Chuvieco, E., 2015. Comparing the accuracies of remote sensing global burned area products using stratified random sampling and estimation. Remote Sens. Environ. 160, 114–121. https://doi.org/10.1016/j.rse.2015.01.005.
Pan, S.J., Yang, Q., 2009. A survey on transfer learning. IEEE Trans. Knowl. Data Eng. 22 (10), 1345–1359. https://doi.org/10.1109/TKDE.2009.191.
Panisset, J., DaCamara, C.C., Libonati, R., Peres, L.F., Calado, T.J., Barros, A., 2017. Assigning dates and identifying areas affected by fires in Portugal based on MODIS data. Anais da Academia Brasileira de Ciências 89 (3), 1487–1501. https://doi.org/10.1590/0001-3765201720160707.
Patz, J.A., Engelberg, D., Last, J., 2000. The effects of changing weather on public health. Annu. Rev. Public Health 21 (1), 271–307. https://doi.org/10.1146/annurev.publhealth.21.1.271.
Pelletier, C., Webb, G.I., Petitjean, F., 2019. Temporal convolutional neural network for the classification of satellite image time series. Remote Sens. 11 (5), 523. https://doi.org/10.3390/rs11050523.
Pereira, A., Pereira, J., Libonati, R., Oom, D., Setzer, A., Morelli, F., Machado-Silva, F., De Carvalho, L.M.T., 2017. Burned area mapping in the Brazilian Savanna using a one-class support vector machine trained by active fires. Remote Sens. 9 (11), 1161. https://doi.org/10.3390/rs9111161.
Perez, L., Wang, J., 2017. The effectiveness of data augmentation in image classification using deep learning. arXiv preprint arXiv:1712.04621.
Pinto, M.M., Hurduc, A., Trigo, R.M., Trigo, I.F., DaCamara, C.C., 2018. The extreme weather conditions behind the destructive fires of June and October 2017 in Portugal. In: Advances in forest fire research 2018. Coimbra, Portugal, 12–16 November, pp. 138–145. https://doi.org/10.14195/978-989-26-16-506_13.
Rego, F.C., Fernandes, P., Silva, J.S., Azevedo, J., Moura, J.M., Oliveira, E., Cortes, R., Viegas, D.X., Caldeira, D., Santos, F.D., 2019. Avaliação do Incêndio de Monchique. Technical Report. Observatório Técnico Independente, Assembleia da República, Lisboa, Portugal.
Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., 2019. Deep learning and process understanding for data-driven Earth system science. Nature 566 (7743), 195. https://doi.org/10.1038/s41586-019-0912-1.
Rodrigues, J.A., Libonati, R., Pereira, A.A., Nogueira, J.M., Santos, F.L., Peres, L.F., Rosa, A.S., Schroeder, W., Pereira, J.M.C., Giglio, L., Trigo, I.F., Setzer, A.W., 2019. How well do global burned area products represent fire patterns in the Brazilian Savannas biome? An accuracy assessment of the MCD64 collections. Int. J. Appl. Earth Obs. Geoinf. 78, 318–331. https://doi.org/10.1016/j.jag.2019.02.010.
Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: convolutional networks for biomedical image segmentation. In: Navab, N., Hornegger, J., Wells, W., Frangi, A. (Eds.), Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. MICCAI 2015. Lecture Notes in Computer Science. Springer, Cham. https://doi.org/10.1007/978-3-319-24574-4_28.
Roteta, E., Bastarrika, A., Padilla, M., Storm, T., Chuvieco, E., 2019. Development of a Sentinel-2 burned area algorithm: generation of a small fire database for sub-Saharan Africa. Remote Sens. Environ. 222, 1–17. https://doi.org/10.1016/j.rse.2018.12.011.
Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors. Cogn. Model. 5 (3), 1. https://doi.org/10.1038/323533a0.
Rußwurm, M., Körner, M., 2018. Multi-temporal land cover classification with sequential recurrent encoders. ISPRS Int. J. Geo-Inf. 7 (4), 129. https://doi.org/10.3390/ijgi7040129.
San-Miguel-Ayanz, J., Schulte, E., Schmuck, G., Camia, A., Strobl, P., Liberta, G., Giovando, C., Boca, R., Sedano, F., Kempeneers, P., McInerney, D., Withmore, C., Oliveira, S.S., Rodrigues, M., Durrant, T., Corti, P., Oehler, F., Vilar, L., Amatulli, G., 2012. Comprehensive monitoring of wildfires in Europe: the European forest fire information system (EFFIS). In: Approaches to Managing Disaster-Assessing Hazards, Emergencies and Disaster Impacts. IntechOpen. https://doi.org/10.5772/28441.
Schroeder, W., Oliva, P., Giglio, L., Csiszar, I.A., 2014. The New VIIRS 375 m active fire detection data product: algorithm description and initial assessment. Remote Sens. Environ. 143, 85–96. https://doi.org/10.1016/j.rse.2013.12.008.
Scott, G.J., England, M.R., Starms, W.A., Marcum, R.A., Davis, C.H., 2017. Training deep convolutional neural networks for land–cover classification of high-resolution imagery. IEEE Geosci. Remote Sens. Lett. 14 (4), 549–553. https://doi.org/10.1109/LGRS.2017.2657778.
Smith, L. N., 2018. A disciplined approach to neural network hyper-parameters: Part 1–Learning rate, batch size, momentum, and weight decay. arXiv preprint arXiv:1803.09820.
Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. J. Mach. Lear. Res. 15 (1), 1929–1958.
Stroppiana, D., Bordogna, G., Carrara, P., Boschetti, M., Boschetti, L., Brivio, P.A., 2012. A method for extracting burned areas from Landsat TM/ETM+ images by soft aggregation of multiple Spectral Indices and a region growing algorithm. ISPRS J. Photogramm. Remote Sens. 69, 88–102. https://doi.org/10.1016/j.isprsjprs.2012.03.001.
Sutskever, I., Vinyals, O., Le, Q.V., 2014. Sequence to sequence learning with neural networks. In: Advances in Neural Information Processing Systems, Montreal, Canada, 8–13 December, pp. 3104–3112.
Trigo, R.M., Pereira, J.M., Pereira, M.G., Mota, B., Calado, T.J., DaCamara, C.C., Santo, F.E., 2006. Atmospheric conditions associated with the exceptional fire season of 2003 in Portugal. Int. J. Climatol. 26 (13), 1741–1757. https://doi.org/10.1002/joc.1333.
Van Der Werf, G.R., Randerson, J.T., Giglio, L., Van Leeuwen, T.T., Chen, Y., Rogers,

B.M., Mu, M., van Marle, M.J.E., Morton, D.C., Collatz, G.J., Yokelson, R.J., Kasibhatla, P.S., 2017. Global fire emissions estimates during 1997–2016. Earth Syst. Sci. Data 9, 697–720. https://doi.org/10.5194/essd-9-697-2017.

Viegas, D.X., 2018. Wildfires in Portugal. Fire Res. 2 (1). https://doi.org/10.4081/fire.2018.52.

Wieland, M., Li, Y., Martinis, S., 2019. Multi-sensor cloud and cloud shadow segmentation with a convolutional neural network. Remote Sens. Environ. 230, 111203. https://doi.org/10.1016/j.rse.2019.05.022.

Young, A.M., Higuera, P.E., Duffy, P.A., Hu, F.S., 2017. Climatic thresholds shape northern high-latitude fire regimes and imply vulnerability to future climate change. Ecography 40 (5), 606–617. https://doi.org/10.1111/ecog.02205.

Zhang, L., Zhang, L., Du, B., 2016. Deep learning for remote sensing data: a technical tutorial on the state of the art. IEEE Geosci. Remote Sens. Mag. 4 (2), 22–40. https://doi.org/10.1109/MGRS.2016.2540798.

Zhang, Q., Yuan, Q., Zeng, C., Li, X., Wei, Y., 2018. Missing data reconstruction in remote sensing image with a unified spatial–temporal–spectral deep convolutional neural network. IEEE Trans. Geosci. Remote Sens. 56 (8), 4274–4288. https://doi.org/10.1109/TGRS.2018.2810208.

Zhong, Z., Zheng, L., Kang, G., Li, S., Yang, Y., 2017. Random erasing data augmentation. arXiv preprint arXiv:1708.04896.

Zhu, X.X., Tuia, D., Mou, L., Xia, G.S., Zhang, L., Xu, F., Fraundorfer, F., 2017. Deep learning in remote sensing: a comprehensive review and list of resources. IEEE Geosci. Remote Sens. Mag. 5 (4), 8–36. https://doi.org/10.1109/MGRS.2017.2762307.